

Def.

In a Ring  $R$ , an ideal  $M \subset R$  is called a Maximal ideal if:  
there is no ideal  $N$  such that  $M \subsetneq N \subsetneq R$ .

Equivalently, if an ideal  $N \subseteq R$  satisfies  $M \subseteq N$ , then either  $N=M$  or  $N=R$ .

Def.

In a Commutative Ring  $R$ , an ideal  $P \subset R$  is called a Prime ideal if:  
for any  $a, b \in R$ , if  $ab \in P$ , then either  $a \in P$  or  $b \in P$ .

Equivalently, if  $(ab) \in P$ , then  $(a) \subseteq P$  or  $(b) \subseteq P$ .

Prop.

Let  $R$  be a Commutative Ring, and  $M, P \subsetneq R$  be Proper ideals.

$M$  is a Maximal ideal if and only if  $R/M$  is a field

$P$  is a Prime ideal if and only if  $R/P$  is an integral domain.

Proof. To show the first assertion, notice that:

An ideal  $I \subseteq R$  contains a unit if and only if  $I=R$ , because:

Given  $r \in R$ ,  $r = r \cdot 1 = r \cdot v^{-1} \cdot v \in I$ . And, the converse is trivial, since  $1 \in R = I$ .  
And, every non-zero element of field is unit, so

$F$  is field if and only if if  $I \subseteq F$  is an ideal, then either  $I=\{0\}$  or  $I=F$ .

Now,  $M \subsetneq R$  is maximal ideal if and only if there is no ideal  $N$  s.t.  $M \subsetneq N \subsetneq R$ .

4<sup>th</sup> iso. theorem  $\rightarrow$  if and only if there is no non-trivial proper ideal of  $R/M$   
if and only if  $R/M$  is a field.

And,  $P \subsetneq R$  is prime ideal if and only if  $ab \in P \Rightarrow a \in P$  or  $b \in P$

if and only if  $ab + P = P \Rightarrow a + P = P$  or  $b + P = P$ .

if and only if  $\overline{a} \cdot \overline{b} = 0 \Rightarrow \overline{a} = 0$  or  $\overline{b} = 0$

if and only if  $\overline{P} = R/P$  has no zero divisor

So proved.  $\square$

